

BRIAN (YUAN-AN) LIU

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GEN AI & CYBERSECURITY SPECIALIST

Strategic technologist with 10+ years of experience delivering scalable cloud architectures and enterprise-grade Generative AI solutions. Combines deep Senior Staff Software Engineering leadership (Zscaler, AWS, F5) with dual expertise in AI innovation and cybersecurity to align business objectives with high-impact technical execution. Expert in designing production-ready LLM workflows (RAG, agentic systems, vLLM/SGLang execution engines), optimizing high-throughput cloud infrastructure, and embedding zero-trust and kernel-level security protections. Proven track record driving the evolution from manual engineering practices to automated, agentic workflows and leading high-level system architecture that drives business growth, reliability, and security resilience.

EDUCATION

ILLINOIS INSTITUTE OF TECHNOLOGY, Chicago, IL

Master of Science, Computer Science

Aug 2016 - May 2018

Association: IIT-Security Lab

CHUNG YUAN CHRISTIAN UNIVERSITY, Taoyuan, Taiwan

Bachelor of Science, Computer Science

Sep 2011 - Jun 2015

Association: CYCU-Software Engineering Lab

SKILLS

- **Languages:** Go, Python, C++, C, Shell Script, TypeScript, Java
- **Cloud & Infrastructure:** AWS, Azure, GCP, Multi-Cloud, CDK, Terraform (IaC), Kubernetes, Docker, Helm, Nginx/traefik, Bedrock, SDN, OVS, eBPF, GNU toolchain, BMaaS, Distributed/Embedded Systems
- **AI/ML/Inference:** Gen AI, LLM, ML, RAG, NLP, RAGAS, Agentic AI, LangChain, LangGraph, Hugging Face, GPT, Claude, Gemini, LLaMA, TensorFlow, PyTorch, Fine-Tuning(LoRA/ QLoRA), SGLang, vLLM, Spec Driven Development (SDD), Local/Edge LLM, Quantization (AWQ/ GGUF)
- **Networking/Protocols/Security:** 5G, gRPC, NSH, SFC, VLAN, VxLAN, TCP/IP, REST, SOAP, P2P, ZMQ, SRv6, Zero Trust (mTLS/TLS), CTEM, SIEM, OWASP, SQL injection, prompt injection, header injection, Kernel & Application Security (memfd_create, XSS, CSP and more)
- **Data & Services:** DynamoDB, PostgreSQL, MySQL, PostgresSQL, OpenSearch/ElasticSearch, Git, Kafka, Kinesis, Protobuf, vim, Telemetry, Prometheus, Temporal, Atlas

EXPERIENCE

ZSCALER, INC. San Jose (Remote), CA

Nov 2024 - present

Senior Staff Software Development Engineer (2026 - present)

Senior Software Development Engineer (2024 - 2026)

- Leading technical solution architecture and implementation for a cloud-native security CTEM service utilizing the Zscaler data fabric for high-availability and strategic customer integration that mitigate OWASP Top 10 (Web & LLM) vulnerabilities and enforce kernel-level runtime security isolation (mitigating memfd_create risks and container escapes).
- Optimizing cloud-native infrastructure deployment costs through detailed cost analysis and capacity planning, resulting in projected revenue models for emerging AI/ML service offerings.
- Engineering and deploying production-grade AI workflows for real-time threat log analysis using RAG-based agents, incorporating diverse LLMs (cloud, on-prem, self-hosted) and continuous model training pipelines.
- Designing and scaling core Zscaler Internet Access (ZIA) inventory services utilizing container orchestration (Kubernetes and Docker) to ensure high throughput and scalability for global cloud services.
- Spearheading advanced R&D initiatives across AI and Cybersecurity to define the technical roadmap and future cloud-native security offerings, ensuring architectural alignment with market needs.
- Inventor and author on multiple Generative AI, ML, and cybersecurity patents, establishing technical thought leadership within the organization and the broader security industry.
- Driving internal engineering adoption and defined best practices for integrating cutting-edge AI/ML technologies, including active contribution to relevant open-source AI projects.
- Accelerated through a double-promotion in recognition of high-impact technical leadership, strategic architecture contributions, and cross-functional delivery.

AMAZON WEB SERVICES, INC. (AWS), Sunnyvale, CA

Nov 2019 - Nov 2024

Software Development Engineer 2 (2022 - 2024)

Software Development Engineer 1 (2019 - 2022)

- Architected and developed AI-powered cloud security services using Generative AI, establishing technical thought leadership across ML and cloud security patents.
- Provided technical leadership for engineering teams, driving business planning, defining product requirements with key stakeholders, and delivering high-impact solutions for large-scale product launches.
- Optimized critical infrastructure cost-efficiency through analysis and collaborated with Product Management on revenue projections for new AWS network service offerings.
- Led end-to-end development of critical AWS network services (Private 5G, VPC, Site2Site VPN, Telco Network Builder) and managed hardware acceleration integration (Outposts, EC2, GPUs) in physical data centers.
- Engineered high-throughput data streaming solutions for AWS Aurora, developing an advanced BinLog manipulation library to ensure low-latency replication and consistency.
- Enhanced service reliability and operational excellence by establishing automation pipelines and leading Root Cause Analysis (RCA) efforts for critical AWS services.

F5 NETWORKS, INC, San Jose, CA

Jul 2018 – Nov 2019

Software Engineer 2 (2019 - 2019)

Software Engineer 1 (2018 - 2019)

- Built policy enforcement manager, service function chaining, and modular solutions for F5 BIG-IP.
- Integrated application delivery networking into CI/CD, conducted SDN controller and RFC feasibility studies.
- Collaborated cross-functionally to deliver innovative products, provided technical leadership and mentoring.

KYPHER, LLC, St. Louis, MO

May 2017 - Aug 2017

Application Developer Intern

- Full Stack Developer for Kypher™ HIPAA Compliant Messaging application.

R.O.C. NAVY, FFG-934, Taiwan

Jul 2015 - Jun 2016

Sonar Technician Seaman (Mandatory Military Service)

- Operated and maintained sonar and underwater fire control systems.

MOREMOTE, INC., Taipei, Taiwan

Jul 2014 - Jun 2015

Software Engineer

- Developed applications for two IP-Camera projects (BBCam™ and Ability HomeCam™).
- Implemented microservices with containerization, REST, Kafka with P2P technology.
- Performed strategic assessments, feasibility studies, software release maintenance.
- Worked directly with OEM clients to provide technical consultation and solution delivery.

PUBLICATION

- **A Distributed Virtual Time on Embedded System for Evaluating Cyber-Physical Systems** (ACM SIGSIM-PADS 2019)
(*Best Paper Award*)
- **A Distributed Embedded Linux System Smart Grid Testbed** (GCASR 2018)

PATENT

- **Self-healing and resiliency in radio-based networks using a community model.** Dec 2022
US 11,888,701 B1, Issued Jan 2024
- **Enhancing availability of radio-based applications using multiple compute instances and virtualized network function accelerators at cloud edge** Aug 2022
US 11,937,103 B1, Issued Mar 2024
- **Hybrid Automotive Cybersecurity Monitoring Service** Feb 2025
US 2026/0247154 A1, Pending Aug 2026
- **Confidential(AI/ML Entropy-weighted Clustering)** Jun 2026
Patent pending
- **Confidential(AI Evidence-Gated HA)** Jun 2026
Patent pending
- **Confidential(AI CTEM)** Oct 2025
Patent pending
- **Confidential(AI Threat Detection)** May 2025
Patent pending
- **Confidential** Dec 2022
Patent pending